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The 2D digraph-based NoCs: attractive alternatives to the 2D mesh NoCs

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Abstract This paper proposes two-dimensional directed graphs (or digraphs for short) as a promising alternative to the popular 2D mesh topology for networks-on-chip (NoCs). Mesh is the most popular topology for the NoCs, mainly due to its suitability for on-chip implementation and low cost. However, the fact that a digraph offers a lower diameter than its equivalent linear array of equal cost motivated us to evaluate digraphs as the underlying topology of NoCs. This paper introduces a family of NoC topologies based on three well-known digraphs, namely de Bruijn, shuffle-exchange, and Kautz. We study topological properties of the proposed topologies. We show that the proposed digraph-based topologies have several attractive features including constant node degree, low diameter and cost, and low zero load latency which result in superior performance over the mesh. We introduce a deadlock-free routing algorithm for the proposed NoC topologies and compare NoCs employing the proposed topologies and the mesh topology in terms of power consumption and

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performance. Simulation results also reveal that the proposed NoC topologies offer higher performance and consume lower power than the mesh NoC.

Keywords Network-on-Chip (NoC) · Mesh · Digraph · De Bruijn network · Shuffle-exchange network · Kautz network · Routing · Power consumption · Performance · Simulation

1 Introduction

Network-on-Chip (NoC) is a promising communication paradigm for multiprocessor systems-on-chip (SoCs). This communication paradigm was inspired from packet-based communication networks and aims to overcome the performance and scalability problems of the shared buses and point-to-point dedicated links in multi-core SoCs [1]. Unlike traditional interconnection networks, power consumption is a first-order constraint in NoC design [2]. As a result, not only should the designer optimize the NoC for delay (as in traditional networks), but also for power consumption.

The choice of network topology is an important issue in designing a NoC. Different topologies provide different average inter-node distances, total wiring lengths, and communication flow distributions. These characteristics mainly determine the power consumption and average packet latency [3].

The mesh topology is the most popular topology used in today's regular tile-based NoCs. It is well known that the mesh topology is regular, has low cost, and consumes low power. Despite such favorable advantages for on-chip implementation, some packets may suffer from long latencies due to lack of short paths between distant nodes in a mesh NoC. A number of previous works tried to tackle this shortcoming, by adding some application-specific links between distant nodes in the mesh [4], bypassing some intermediate nodes by inserting express channels [5], or introducing new topologies with lower diameters [6, 7].

Directed graphs (or digraphs) have interesting properties when compared to the linear array topology (which is the main building block of a 2D mesh). They have the same number of links a linear array has; they also have logarithmic diameters, and hence can perform better than a linear array network with a linear diameter. Such important advantages of digraphs motivated us to consider them for designing 2D NoCs. To this end, we have extended the use of a digraph in a 2D grid structure where nodes of each row and each column form a digraph. In other words, we replace the linear arrays in the rows and columns of a 2D mesh with the same-sized digraphs.

In this paper, we further study and compare the performance and power consumption of three 2D NoC topologies, the shuffle-exchange-mesh [8], Bruijn-mesh [9], and Kautz-mesh [10], respectively based on three well-known digraphs, the de Bruijn, shuffle-exchange, and Kautz digraphs, with the most known 2D mesh NoC topology. This paper extends the prior works in [8–10] by introducing more effective routing algorithms for the proposed 2D NoC topologies and comparing them by means of detailed simulation experiments. More precisely, we compare different NoCs (employing 2D mesh, shuffle-exchange-mesh, Bruijn-mesh, and Kautz-mesh) in terms of network latency, power consumption, and used area, and under different working

conditions. Simulation results show that the average message latency and power consumption of the proposed networks are improved compared to the equivalent mesh NoCs.

The rest of the paper is organized as follows. Section 2 provides a background on digraphs. Section 3 describes the 2D digraph-based NoC topologies along with their routing algorithms. Section 4 extends the mesh-based digraph topologies to the torus-based digraphs. Section 5 compares the proposed NoC topologies and the equivalent mesh NoC topology in terms of performance, power consumption, and area. Simulation results are discussed in Sect. 6, and finally, Sect. 7 concludes the paper.

2 Digraphs and their routing algorithms

Graphs can be effectively used to model network-based systems. A directed graph (or digraph) is then a suitable model for a network with unidirectional links. In this section, we briefly review on some well-known digraphs and their routing algorithms. In what follows, we interchangeably use *graph* and *network*.

2.1 The shuffle-exchange network

The traditional shuffle-exchange network was first proposed in [12]. It is a popular interconnection topology for multiprocessors and multicomputers due to its scalability and distributed self-routing capability [13]. Several researchers have studied the topological properties [12] and efficient VLSI layout [14] of the shuffle-exchange network.

In a 2^n-node shuffle-exchange network, each node is identified by a unique n-bit binary address. Two nodes are directly connected (via *exchange* bidirectional link which is equivalent to two unidirectional links) to each other if their addresses differ in their least significant bits. A node is connected to another node (by *shuffle* unidirectional link) whose address is a one-bit cyclic shift of its address. Movement of data between the adjacent nodes via shuffle and exchange links is respectively called shuffle and exchange operations. Messages generated at source nodes can be communicated between network nodes using shuffle and exchange operations until reaching their destination nodes.

Shuffle and exchange operations are defined on an n-bit address pattern $\mathbf{A} = A_{n-1}A_{n-2}\ldots A_1A_0$ as

$$\begin{aligned}\text{Shuffle: } &(A_{n-1}A_{n-2}\ldots A_1A_0) \rightarrow A_{n-2}\ldots A_1A_0A_{n-1}\\ \text{Exchange: } &(A_{n-1}A_{n-2}\ldots A_1A_0) \rightarrow A_{n-1}A_{n-2}\ldots A_1A_0\end{aligned}$$

Each node has two outgoing connections to other nodes and two incoming connections from other nodes. Since these connections are unidirectional, the degree of the network is the same as the one-dimensional mesh or the linear array network. The diameter of a shuffle-exchange network of size $N(=2^n)$ is $2n-1$, which is the minimum distance between the two end nodes, i.e. nodes 0 and 2^n-1. Figure 1 shows an 8-node shuffle-exchange network.

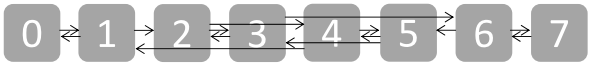


Fig. 1 An 8-node shuffle-exchange network

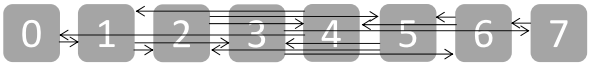


Fig. 2 The de Bruijn network with 8 nodes

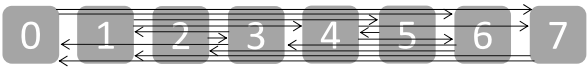


Fig. 3 The Kautz network with 8 nodes

2.2 The de Bruijn topology

The de Bruijn topology is another well-known digraph proposed [15] as an efficient topology for multiprocessors. Samathan and Pradhan [16] showed that de Bruijn networks are also favored for VLSI implementation, and several other researchers have studied topological properties, routing algorithms, efficient VLSI layout and other important aspects of de Bruijn networks [11, 16–18].

In a binary de Bruijn digraph of 2^n nodes, each node $\mathbf{A} = A_{n-1}A_{n-2}\ldots A_1A_0$ has direct connections to nodes $A_{n-2}\ldots A_1A_00$ and $A_{n-2}\ldots A_1A_01$. Both in-degree and out-degree of a node are 2, making a total node degree equal to that in a linear array network. These connections are defined by shuffle (rotate left by one bit) and shuffle-exchange (rotate left by one bit and complement the least significant bit) operations on node address [19].

The diameter of a de Bruijn graph is equal to n. The logarithmic diameter and the simple routing mechanism in de Bruijn networks are the two most desirable properties of this family of digraph networks. An example of the de Bruijn topology is illustrated in Fig. 2.

2.3 The generalized Kautz digraph

A class of digraphs that generalize Kautz digraphs [20, 21] is defined in [22]. Suppose that the vertices are numbered with integers modulo N (N being the number of nodes). Assuming an out-degree of d, then a vertex v is joined to vertices $u = -dv - i \pmod N$, for $i = 1, 2, \ldots, d$. The diameter of the resulting digraph is at most $\lceil \log_d N \rceil$ and if $N = d^n + d^{n-k}$ (for a positive odd integer k), the diameter becomes n.

Based on the above definition, an N-node *binary* Kautz digraph is then defined as follows. Node v, with a linear address $0 \ldots N-1$, is joined to vertices with integer address $-2v - 1 \pmod N$ and $-2v - 2 \pmod N$. The diameter of such a network is $\lceil \log_2 N \rceil$. An 8-node binary Kautz digraph is depicted in Fig. 3.

2.4 Routing in digraphs

Several routing algorithms have been proposed in the literature for the shuffle-exchange and de Bruijn networks [17, 23]. The digraphs considered in this paper

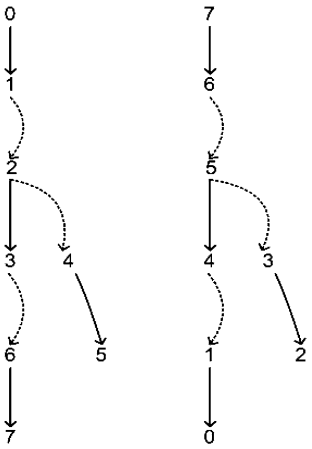


Fig. 4 Increasing (*left*) and decreasing (*right*) sub-graphs for an 8-node shuffle-exchange network

have some topological similarities that can be used for introducing a common routing scheme for the proposed digraph-based 2D mesh NoCs (in Sect. 3.4). In this section, we review on the routing algorithms proposed in the literature for digraphs.

Dally [24] presented a routing algorithm for shuffle-exchange networks that routes the packets from the source node toward the destination node by changing the address one bit at each hop. It starts from the most significant bit of the n-bit source address. At the ith step of the algorithm, the $(n-i)$th bit of the destination address is compared to the LSB (least significant bit) of the current node address. If equal, the message is routed over the shuffle link keeping the bit unchanged and rotating the address bit pattern. Otherwise, the message is routed over the exchange link to make the two bits equal. This algorithm involves a maximum of $2n$ communication steps between adjacent nodes along the path from the source node to the destination node. However, this algorithm cannot always find the shortest path between source-destination pairs [24]. In order to be deadlock-free, this algorithm requires n virtual channels per physical channel and the message uses the ith virtual channel at the $(n-i)$th step. Since the virtual channel selection is done in decreasing order of virtual channel number, the dependency graph of the involved virtual channels is acyclic, thus the routing is deadlock-free [24].

Ganesan [18] proposed a similar routing algorithm for de Bruijn networks. The only difference between his scheme and Dally's algorithm is that: if the two compared bits (in Dally's algorithm) are equal, the message is routed over the shuffle link, to keep the bit unchanged and rotate the address pattern. Otherwise, the message is routed over the shuffle-exchange link to make the two bits equal and also rotate the address pattern. This algorithm involves a maximum of n inter-node communication steps.

Park et al. [17, 23] improved Dally's routing algorithm using a lower number of virtual channels per physical channel. Park represented the shuffle-exchange and de Bruijn as two sub-graphs, namely *increasing* and *decreasing* sub-graphs, as shown in Figs. 4 and 5. In the ith step, the MSB of the current node address pattern is compared to the $(n-i)$th bit of the destination node address pattern; if they are equal, shuffle cycle is used; otherwise, the data is forwarded by an exchange operation (shuffle-

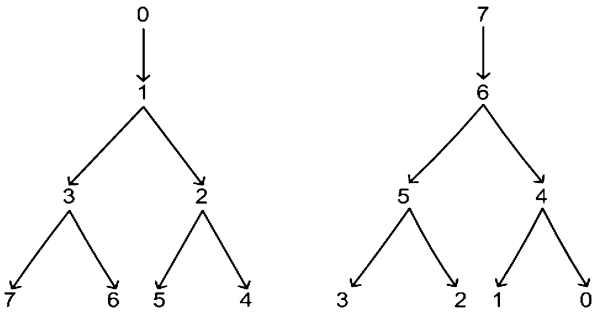


Fig. 5 Increasing (*left*) and decreasing (*right*) sub-graphs for an 8-node de Bruijn network

exchange operation in the de Bruijn network). If the path switches from the increasing sub-graph to the decreasing sub-graph, the virtual channel number is increased by one. They proved that the mentioned algorithm for the de Bruijn network is deadlock-free [17, 23]. Park [17, 23] showed that for shuffle-exchange and de Bruijn networks of N nodes ($N = 2^n$), this routing scheme requires $n - \lfloor (n-1)/2 \rfloor$ virtual channels to ensure deadlock freedom. For example, for deadlock-free routing in 8-node and 16-node networks, it requires respectively 2 and 3 virtual channels per physical channel.

Bopanna [25] introduced another deadlock-free routing algorithm for de Bruijn network which is minimal and uses two sub-graphs. In his method, the links of the de Bruijn graph are grouped into two classes: 0-edges and 1-edges. A 0-edge (or 1-edge) link connects a node to another node with a 0 (or 1) in the most significant bit position. Bopanna has proved that $\lceil (n+1)/2 \rceil$ virtual channels per physical channel are sufficient for deadlock-free routing. For example, for deadlock-free routing in 8-node and 16-node networks, according to Bopanna's scheme it requires respectively 2 and 3 virtual channels per physical channel.

3 The proposed 2D digraphs NoC

In this section, we introduce the two-dimensional digraphs as some alternative topologies for the popular mesh topology widely used in NoCs. The motivation comes from the facts that digraph networks have logarithmic diameters and their node degree is equal to that of the linear array network (or 1-D mesh).

3.1 The two-dimensional shuffle-exchange-mesh

If in a two-dimensional $2^n \times 2^n$ mesh network, we remove the links and interconnect the nodes in each row and column as a 2^n-node shuffle-exchange digraph, a two-dimensional shuffle-exchange-mesh (or 2-D SEM for short) network is formed. Figure 6 shows a 2-D SEM network with 64 nodes.

Node degree: In a 2-D SEM network, each node has two outgoing and two incoming links in each dimension and, thus, has a total of 8 unidirectional links in two dimensions. Thus, the node degree of the 2-D SEM is equal to that in a traditional 2-D mesh network (i.e. 4 bidirectional links). The node degree of a topology contributes to (and usually acts as the dominant factor of) the network cost.

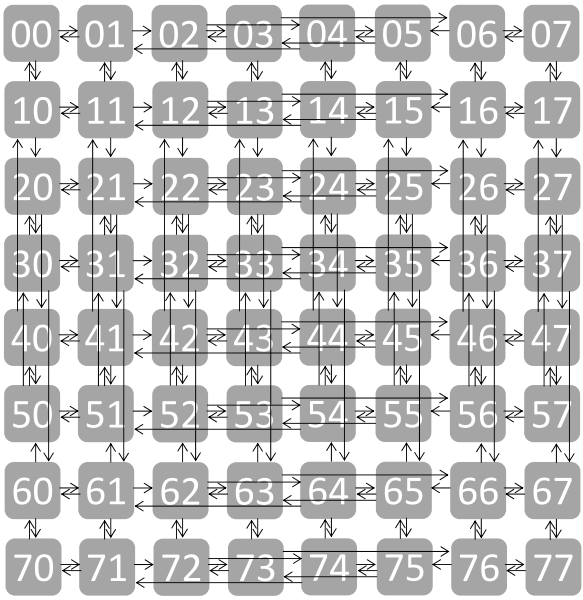


Fig. 6 The 2D SEM with 64 nodes

Diameter: The diameter is the maximum distance between two nodes in the networks [26]. The smaller the diameter of a network is, the less time it takes to send a message from one node to the node farthest away. The network diameter of an N-node 2-D SEM is $4 \log N^{0.5} - 2$, which is smaller than the diameter of the equivalent 2-D mesh with a diameter of $2N^{0.5} - 2$.

Cost: To evaluate the total cost of a network, the node degree and network diameter are used that represent static and dynamic costs of the network. In networks, there is usually a tradeoff between the node degree and network diameter. The cost of a network is defined as the product of the number of network (bidirectional) links and network diameter [27]. So, the cost of an N-node 2-D SEM can be defined to be $2N(4 \log N^{0.5} - 2)$. The cost of an equivalent 2-D mesh is $2N(2N^{0.5} - 2)$.

Bisection width: The bisection width of a network is the minimum number of links that must be removed to divide the networks into halves. A 2^n-node shuffle-exchange network [28] has a bisection width of not less than $2^{n-1}/n$. Thus, the bisection width of an N-node 2-D SEM bisection bandwidth is not less than $N/(2 \log N^{0.5})$. The bisection width of an equivalent 2-D mesh is $N^{0.5}$.

3.2 The two-dimensional de Bruijn-mesh and Kautz-mesh

Interconnecting each row and column in a 2-D mesh network as a de Bruijn digraph results in a two-dimensional de Bruijn-mesh (or 2-D BM for short) network. Each node has two outgoing and two incoming links in each dimension resulting in a total of 8 unidirectional links (or 4 bidirectional links in a 2-D mesh). Figure 7 shows an example of 2-D BM network.

Each node has two outgoing and two incoming links in each dimension. Thus, each node has a total of 8 unidirectional links. The 2-D BM links (similar to 2D SEM) are

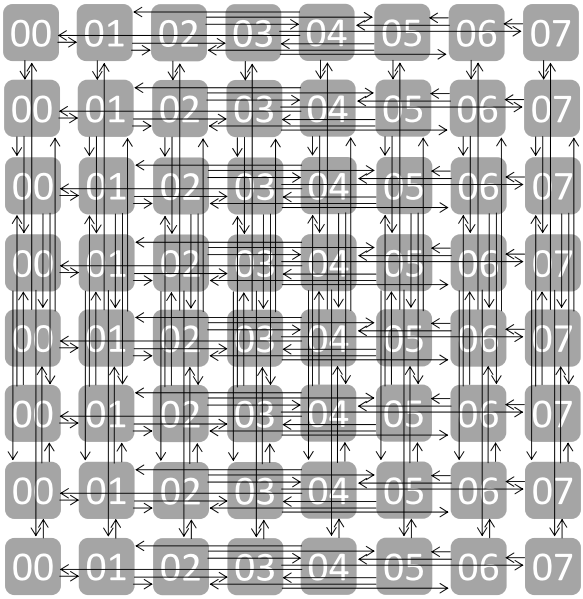


Fig. 7 The 2D DBM with 64 nodes

unidirectional and the 8 unidirectional links are used per node. This is equal to the number of links connected to a node in a mesh (i.e. 4 bidirectional links).

The diameter of a 2-D BM is $2\log N^{0.5}$, where N is the network size. The cost of a 2-D BM, defined as the product of the number of links and diameter of the network, is then equal to $4N\log N^{0.5}$. Considering that the bisection width of a 2^n-node de Bruijn digraph is $2^n/n$ [28], the bisection width of an N-node 2-D BM can be given as $N/\log N^{0.5}$.

The links generated by shuffle operations can connect some distant nodes in corresponding positions of a same-sized mesh, and reduce the distance between two end points of the network. However, by establishing the new links, the direct links between some adjacent nodes (for example 1 to 0, 2 to 1 and 3 to 4 connections in Figs. 2 and 7) are removed and their distance is increased by one hop. Considering all possible source-destination pairs, it can be concluded that the inter-node distance for many pairs is decreased by one or more hops (although it is increased for some pairs) and this can lead to a considerable reduction in the average inter-node distance in the network.

A necklace in a de Bruijn is the set of nodes in which the address of one node is a cyclic shift of the address of another node [27]. The weight of a node is defined as the number of 1s in the node address. A group G_ω is defined as the set of nodes with the same weight ω. In de Bruijn digraphs, the necklaces are placed by the order of their weights. Furthermore, in group G_ω, nodes as 1xx1 are placed near group $G_{\omega-1}$ and nodes as 0xx0 are placed near group $G_{\omega+1}$ [27].

A more efficient layout for the de Bruijn graph [9] can be used, as shown in Fig. 8. The layout of a 2-D BM network can be found by extending the layout proposed in [9] for two dimensions. With the new layout the total wiring length in the network is decreased. For example, for an 8×8 2-D BM network a reduction of about 25% in

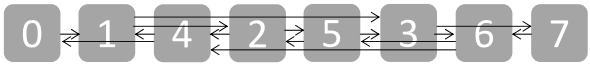


Fig. 8 A better node placement of de Bruijn network of 8 nodes

Table 1 Topological properties of the considered networks

Network topology	Network size	Node degree	Network diameter	Network cost	Network bisection width	Average wire length for an 8 × 8 network
Mesh	$2^n \times 2^n$	4	$2(2^n - 1)$	$2^{2n+2}(2^n - 1)$	2^n	2 mm
2D SEM	$2^n \times 2^n$	4	$2(2n - 1)$	$2^{2n+2}(2n - 1)$	$>2^{2n-1}/n$	2.85 mm
2D DBM	$2^n \times 2^n$	4	$2n$	$2^{2n+2}n$	$2^{2n}/n$	3.42 mm

total wiring can be obtained using the new layout; this reduction can further improve to 50% for a 16 × 16 2-D BM, and even more for larger networks.

A two-dimensional Kautz-mesh network can be defined by removing the normal links of the 2-D mesh network and interconnecting the nodes of each row or column as a generalized binary Kautz digraph (defined above). The resulting network is called two-dimensional Kautz-mesh (or 2-D KM for short). The topology of a $2^n \times 2^n$ 2-D KM topology is isomorphic to the $2^n \times 2^n$ 2-D BM. Thus, all properties studied in the $2^n \times 2^n$ 2-D BM are reserved in the $2^n \times 2^n$ 2-D KM network when the radix of the network is a power of 2. Thus, in the rest of paper, we only consider the 2-D SEM and 2-D BM (as the 2-D KM has similar properties to the same-sized 2-D BM). The obvious advantage of the 2-D KM to the 2-D SEM and 2-D BM, however, is its size variation that can be any arbitrary radix (rather than being a power of 2 in the 2-D SEM and 2-D BM).

3.3 Reducing the effect of long wires

Table 1 compares topological attributes of the proposed 2-D digraph-based mesh networks. One important advantage of the mesh NoC is its short wiring requirements between adjacent nodes. But in NoCs based on the proposed topologies, some internode wires are longer than the wiring length (or wire segment) in the equivalent 2-D mesh NoC. However, long wires may not degrade the NoC operating frequency, since the operating frequency of a NoC is often determined by the longest router pipeline stage. This can be achieved by segmenting long links into regular fixed length links (wire segments in the equivalent mesh NoC) connected by 1-flit buffers [4]. Here, 1-flit buffers provide pipelining over the links while at the same time act as repeaters. By pipelining the transmission of the flits of a message over a long segmented link, the effect of long wires on the performance becomes almost negligible [4]. However, the extra buffering and wiring requirements must be taken into account when calculating the power consumption.

3.4 Routing algorithms in the proposed 2-D digraph-based networks

We adopt the routing methods introduced in [17, 23] (described in the previous section) by Park as a common routing scheme when developing routing algorithms for the proposed digraph-based mesh networks.

```
Algorithm Routing (C, D, Vadd, VC)
Inputs:    Current node C=(XC, YC); Destination node D=(Xd, Yd);
           Current sub-graph indicator Vadd;  // can be increasing or decreasing
           Current virtual channel VC;
Begin

   If (C=D) then return EjectionChannel;       // message is at the destination node

   If (XC#Xd) then
         (PC, g) = Next-Node-X (XC, Xd);     // PC is an output physical channel of X dimension belonged to sub-graph  g
         If Vadd= decreasing and g=increasing then VC++;
         Vadd=g;
         return (PC, VC);
   Endif;

   If (YC#Yd) then
         (PC, g) = Next-Node-Y (YC, Yd);     // PC is an output physical channel of Y dimension belonged to sub-graph  g
         If Vadd= decreasing and g=increasing then VC++;
         Vadd=g;
         return (PC, VC);
   Endif;

End.
```

Fig. 9 The pseudo code of the routing algorithm

Park's algorithm uses virtual channels unevenly where very few packets use all the virtual channels. Numbering virtual channels in an increasing order starting from 0, it uses virtual channel 0 more than other virtual channels. For example, for $N = 16$, some of the source nodes use only virtual channel 0, some nodes use virtual channels 0 and 1, and a few source nodes use virtual channels 0, 1, and 2.

This routing algorithm was revised to make a more balanced use of virtual channels. Here, at each node, the packet has a degree of flexibility in selecting virtual channels [9]. For the same example of $N = 16$ given above, the last group of source nodes (that use 3 virtual channels), virtual channel 0 is selected to inject the message. For the second group, packets start their journey with virtual channel 1, and for the first group (using 1 virtual channel) they start with virtual channel 2. If a message wants to start with virtual channel 2 and it is occupied, it can try virtual channel 1, and if it is also occupied it can try starting with virtual channel 0. Through this method, the virtual channels are used more uniformly that results in a more balanced traffic over network channels. Figure 9 shows the pseudo code of the routing algorithm.

In this paper, we apply the routing algorithm given in Fig. 9. To this end, a message first routes the packet in X dimension using the above-mentioned routing algorithm in such a way that the route length be minimal. When the x-value of the current node and x-value of the destination node addresses become equal, the packet is then routed to the destination by applying the same routing algorithm in the destination columns. Note that routing in the two dimensions cannot generate cyclic dependencies as the base routing algorithm in each dimension is deadlock-free and the packets travel the network in a dimension order. Therefore, the resulted routing algorithm is deadlock-free.

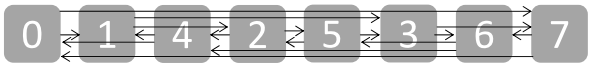


Fig. 10 The Torus-Bruijn with 8 nodes

Table 2 The effect of wraparound to the de Bruijn network

The path in the 8-node de Bruijn network	The new path in the modified de Bruijn network
$0 \rightarrow 1 \rightarrow 3 \rightarrow 7$	$0 \rightarrow 7$
$0 \rightarrow 1 \rightarrow 3 \rightarrow 6$	$0 \rightarrow 7 \rightarrow 6$
$4 \rightarrow 1 \rightarrow 3 \rightarrow 7$	$4 \rightarrow 0 \rightarrow 7$
$4 \rightarrow 1 \rightarrow 3 \rightarrow 6$	$4 \rightarrow 0 \rightarrow 7 \rightarrow 6$
$7 \rightarrow 6 \rightarrow 4 \rightarrow 0$	$7 \rightarrow 0$
$7 \rightarrow 6 \rightarrow 4 \rightarrow 1$	$7 \rightarrow 0 \rightarrow 1$
$3 \rightarrow 6 \rightarrow 4 \rightarrow 0$	$3 \rightarrow 7 \rightarrow 0$
$3 \rightarrow 6 \rightarrow 4 \rightarrow 1$	$3 \rightarrow 7 \rightarrow 0 \rightarrow 1$

4 Torus-based networks

To improve the performance of a NoC employing the proposed digraph-based meshes, a variety of methods can be considered. For example, 3D VLSI technologies can be used for implementing 2-D digraph-based meshes [8, 10].

Also, nodes at the edges of the digraphs can be inter-connected to reduce the diameter and average inter-node distance of the network. This can generate 2-D digraph-based meshes with wraparounds or digraph-based torus networks. For example, in the 2-D BM, we can add wraparound links between edge nodes 0 and $2^n - 1$ to make a 2-D de Bruijn-torus. Furthermore, a shuffle-exchange torus network can be formed likewise by adding wraparound links to the corresponding 2-D SEM.

Figure 10 shows a de Bruijn network of $N = 8$ nodes which is modified by connecting the edge nodes 0 and 7 using two unidirectional wraparound links. Here, a packet residing on nodes 0 and 4 uses the added link to reach nodes 6 or 7. Similarly, nodes 3 and 7 are connected to nodes 0 and 1 through the added links. To move from node 0 to node 6, a message can move to node 7 over the added link using virtual channel 0, and then from node 7 to node 6 using virtual channel 1. The improvements achieved on hop counts (or new paths) using the wraparounds are shown in Table 2 for an 8×8 de Bruijn-mesh.

The routing algorithm proposed for the digraph-based meshes can be simply modified to match the new topology. The number of virtual channels required for deadlock-free routing in such a topology is the same as the original network; here, in addition to aforementioned virtual channel selection policies, after traversing a wraparound link, virtual channel number is increased by one.

5 Experimental results

In this section, the performance and power consumption of the proposed topologies are evaluated and compared to those of the equivalent meshes. The simulation results

were obtained for 8×8 and 16×16 mesh NoCs using XY routing algorithm, and 8×8 and 16×16 2-D SEM and 2-D BM NoCs using the routing algorithm described in Sect. 3.4. As mentioned before, except for their size, which can be selected with more flexibility, the 2-D KM NoCs are similar to other proposed digraph-based NoC topologies. Therefore, we omit the power and performance results of the 2-D KM NoCs for the sake of brevity.

5.1 Simulation environment

To simulate the proposed NoC topologies, an interconnection network simulator that is developed based on POPNET [29] with an embedded latest version of Orion power library [30] was used. Providing detailed power characteristics of the network elements, Orion enables the designers to make rapid power evaluation at the architecture level [32].

The POPNET simulator was modified to mimic the exact operation of the 2-D mesh and the 2-D digraph-based mesh NoCs. The simulator was also customized to support the corresponding routing algorithms for the considered networks. The power consumption can be obtained for each component of the network and for each layer (if implemented using 3D VLSI technology).

For each simulated network, the physical link width was assumed to be 32 bits. The power was calculated based on a NoC with 90 nm technology whose routers operate at 250 MHz. Based on the core size information presented in [31], the side size of each IP core was set to 2 mm, and the length of each wire was set based on the number of cores it passes through.

The message length was assumed to be $M = 32$ and 64 flits and $V = 2$ and 3 virtual channels per physical channel were used. At each node, messages were generated according to a Poisson distribution with an average generation rate of λ messages per cycle. The simulations were performed under uniform, matrix-transpose, and hotspot [26] traffic patterns.

Note that for matrix-transpose traffic load, it was assumed that 30% of the messages generated by a node were of matrix-transpose type (i.e. node (x, y) sends a message to node (y, x)) and the rest of the messages were sent to other nodes uniformly. In hotspot traffic load, a hotspot rate of 16% was assumed, i.e. each node sent 16% of messages to the hotspot node (which is assumed to be node (4,0) in the 8×8 network, and node (8,0) in the 16×16 network) and the rest of messages were sent to other network nodes uniformly.

For each simulation scenario, a total of 500K messages were generated where the first 50K messages were considered for warm up period of the network.

5.2 Simulation results

In Fig. 11, the average message latency is displayed as a function of message generation for the 8×8 2-D mesh, SEM, and BM (with improved layout) NoCs using $V = 2$ virtual channels per physical channel and for two different message sizes $M = 32$ and 64 flits under various traffic patterns.

As shown in Figs. 11a and 11b, under uniform traffic, the 2-D BM network has the smallest average message latency for both message sizes, because the average

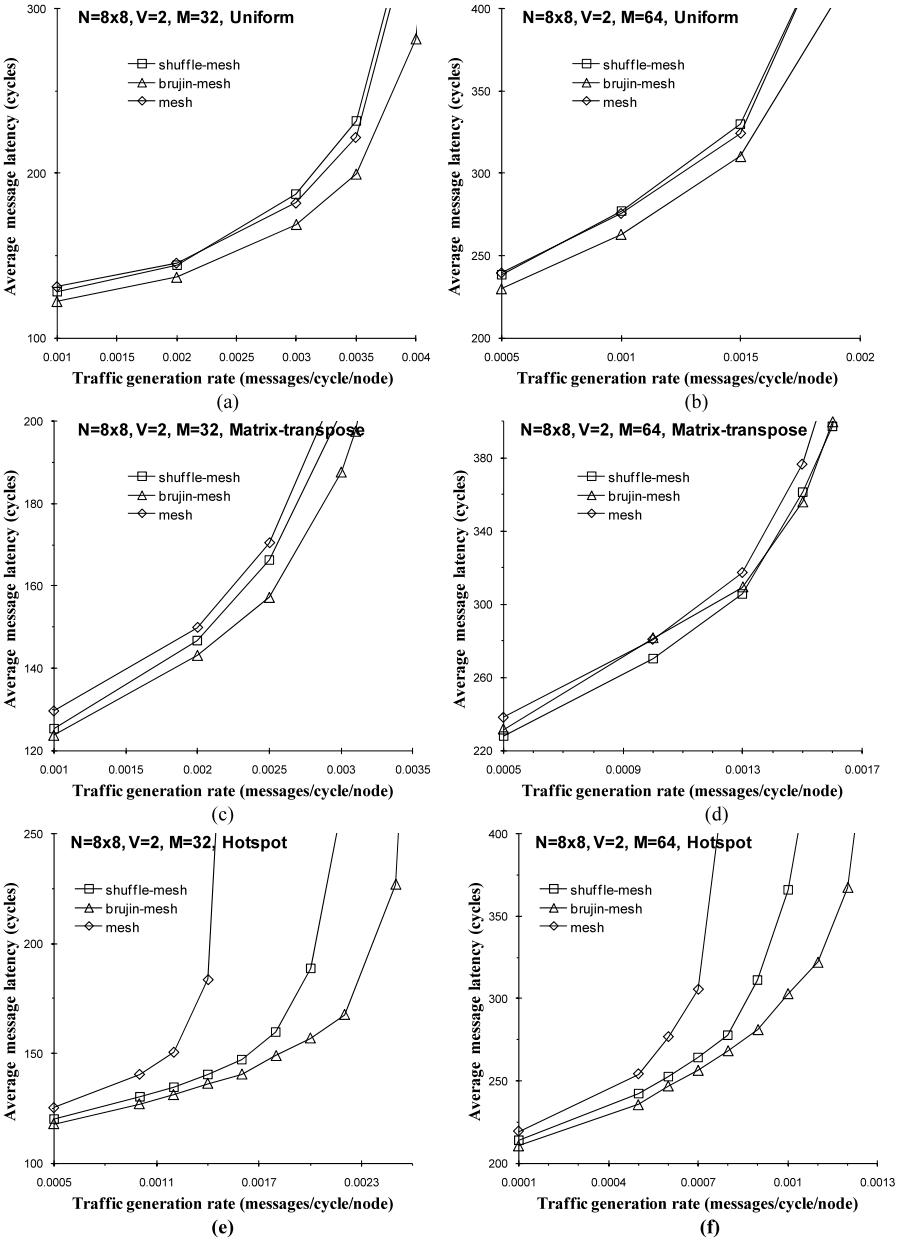


Fig. 11 The average message latency of 8 × 8 mesh and 2D Digraph NoCs using 2 virtual channels per physical channel with 32-flit and 64-flit messages

inter-node distance is logarithmic and lower than that of other equivalent networks. Figures 11c and 11d report similar results for matrix-transpose and Figs. 11e and 11f show the results for hotspot traffic patterns. As can be seen in the figure, the 2-D

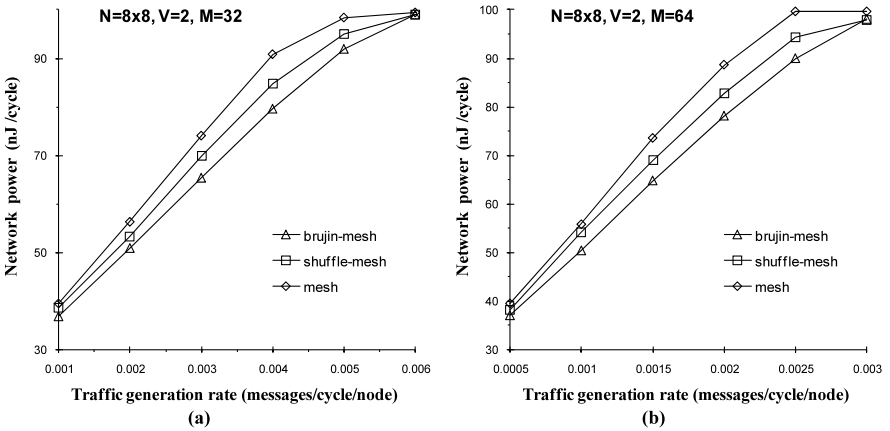


Fig. 12 The power consumption of 8 × 8 mesh and 2D Digraphs NoCs using 2 virtual channels per physical channel with 32-flit and 64-flit messages

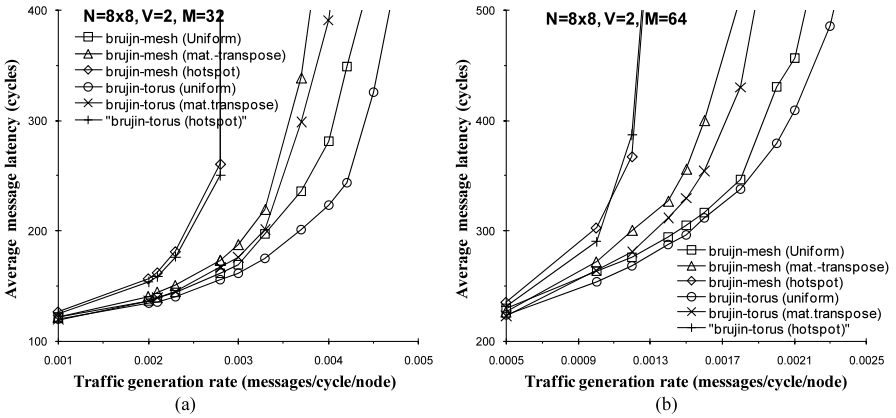


Fig. 13 The average message latency of 8 × 8 torus-Bruijn and 2D DBM NoCs using 2 virtual channels per physical channel with 32-flit and 64-flit messages

BM can better cope with non-uniformity of the network traffic and its performance improvement over its 2-D SEM and 2-D mesh counterparts is more visible under hotspot traffic pattern.

Figure 12 depicts power consumption of the considered networks under uniform traffic. As shown in the figure, the proposed 2-D digraph topologies can effectively reduce the power consumption of the NoC compared to the equivalent mesh topology. The main source of such a noticeable reduction is the less hop counts taken by the messages (on average) and hence saving the power which is consumed in intermediate routers in an equivalent mesh topology.

Figures 13 and 14 compare the performance of the 2-D shuffle-exchange-torus and de Bruijn-torus and their 2-D SEM and 2-D BM counterparts under different traffic patterns. Here, the network has 8 × 8 nodes operating under various traffic patterns,

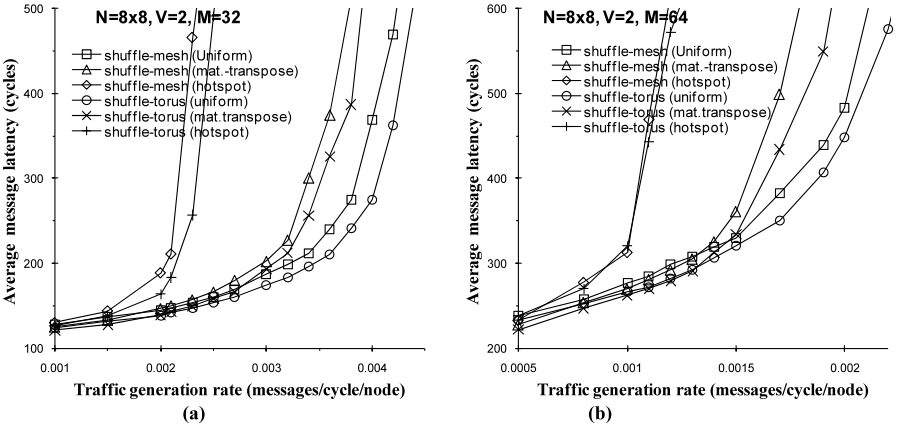


Fig. 14 The average message latency of 8 × 8 torus-shuffle and 2D SEM NoCs using 2 virtual channels per physical channel with 32-flit and 64-flit messages

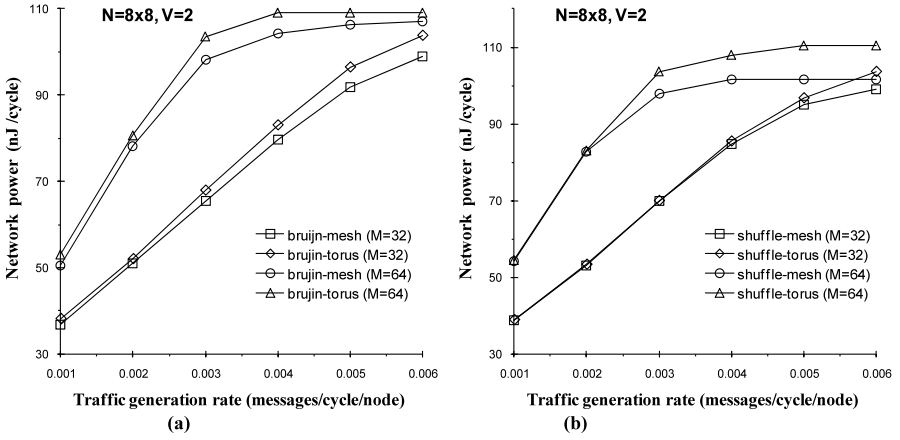


Fig. 15 The power consumption of 8 × 8 torus and 2D Digraphs NoCs using 2 virtual channels per physical channel with 32-flit and 64-flit messages for Bruijn and shuffle-exchange

message lengths $M = 32$ and 64 flits, and $V = 2$ virtual channels per physical channel. In Fig. 13 the average message latency is displayed as a function of the traffic generation rate at each node for the 8 × 8 2-D de Bruijn-mesh and torus networks. Figure 14 shows similar results for the shuffle-exchange-based mesh and torus networks. It is clear that the 2-D digraph-based torus networks achieve a reduction in average message latency with respect to their 2-D digraph-based mesh counterparts for the full range of network loads under various traffic patterns (particularly under uniform traffic).

Figure 15 shows the power consumption of the 2-D digraph-based torus networks and their and 2-D digraph-based mesh counterparts under uniform traffic for message lengths $M = 32$ and 64 flits. It can be seen in the figure that the 2-D torus-based

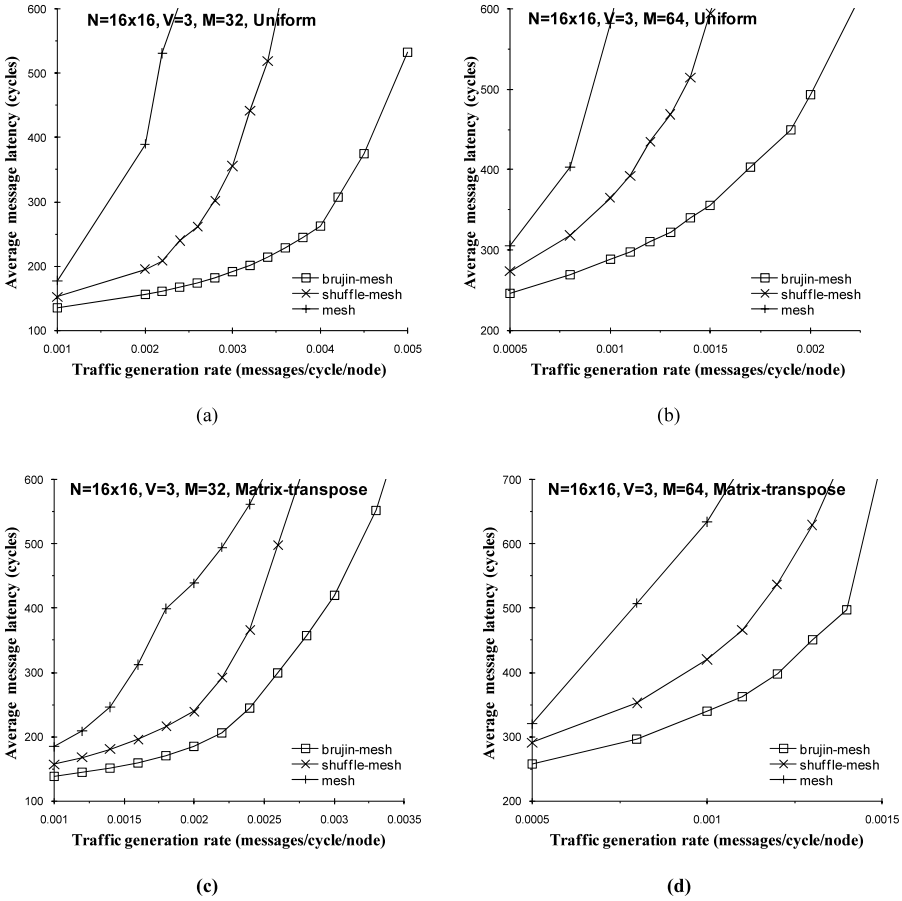


Fig. 16 The average message latency of 16 × 16 mesh and 2D Digraph NoCs using 3 virtual channels per physical channel with 32-flit and 64-flit messages

networks consume more power as a result of their extra links (wraparounds). This trend continues even after the mesh-based networks are saturated where the torus-based networks can still handle more traffic, and thus consuming more power.

To evaluate the scalability of the proposed topologies, Fig. 16 shows the average message latency as a function of message generation rate at each node for larger NoCs, i.e. the 16 × 16 mesh, 2-D SEM and 2-D BM NoCs under uniform and matrix transpose traffic loads, message lengths of $M = 32$ and 64 flits, and $V = 3$ virtual channels per physical channel (note that for 16-node digraph network 3 virtual channels per physical channel must be used for deadlock-free routing). Note that increasing the network size causes earlier saturation in a simple 2-D mesh NoC (compare Figs. 16 and 12). As shown in the figures, again the 2-D BM and 2-D SEM networks attain a reduction in message latency with respect to the 2-D mesh network for the full range of network loads.

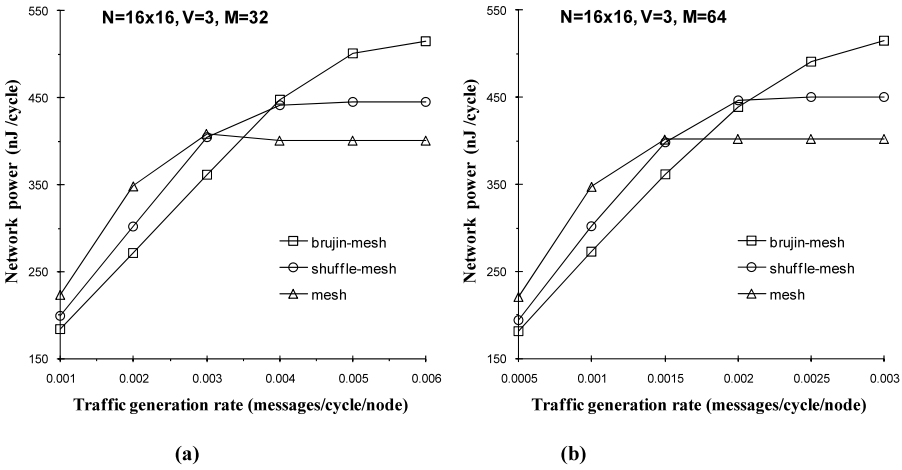


Fig. 17 The power consumption of 16 × 16 mesh and 2D Digraph NoCs using 3 virtual channels per physical channel with 32-flit and 64-flit messages

Figure 17 depicts the power consumption of the 16 × 16 2-D mesh, 2-D BM and 2-D SEM under uniform traffic. Note that the proposed topologies have reduced the power consumption of the NoC before reaching the saturation point. Obviously, handling more traffic load (after the point that the 2-D mesh is saturated) requires more power. Note that when the mesh network approaches its saturation region, the 2-D digraph-based networks can still handle the traffic effectively and the saturation point for them is higher than that in the equivalent mesh network.

5.3 Area estimation

The area overhead due to the additional inter-router wires was analyzed by calculating the number of channels in a NoC. An $n \times n$ 2-D mesh has $2n(n-1)$ channels. The 2-D digraph-based NoCs have the same number of channels although some links use longer wires. To evaluate the imposed area overhead, we estimated the area of the routers and links using Orion 2.0 tool [32]. The area of the router was calculated in 90 nm technology.

In the analysis, the lengths of input and output network interface buffers were considered as large as 64 flits. This was a modest size for the network interface queues (larger queues are used in some NoC implementations, e.g. in [33]).

In Table 3, the area overhead of a digraph-based NoC is evaluated for 8 × 8 and 16 × 16 networks. The results show that in an 8 × 8 mesh the total area of the links and routers are 0.0366 mm^2 and 0.2742 mm^2, respectively. Based on these area estimations, the area of the network part of the 2-D BM network shows a 22% increase compared to the 2-D mesh NoC of equal size. Considering 2 mm × 2 mm processing elements, the increase in the entire chip area is about 2%. The area of the network part of an 8 × 8 2-D SEM network shows a 13% increase compared to a simple 2-D mesh NoC. This increases the entire chip area by less than 1.3%.

Table 3 Area comparison

Network	Link area (mm^2)	Router area (mm^2)	Added area with respect to the 2-D Mesh network (percentage)	Added area with respect to the whole chip (IP cores and a 2-D Mesh network)
64-node 2-D Mesh	0.0366	0.2742	0	0
64-node 2-D SEM	0.0523	0.2742	13.48	1.24
64-node 2-D BM	0.0627	0.2742	22.75	2.08
256-node 2-D Mesh	0.0366	0.2969	0	0
256-node 2-D SEM	0.0879	0.2969	44.27	4.33
256-node 2-D BM	0.0939	0.2969	49.52	4.85

In Table 3, the area of a 16×16 2-D SEM network shows a 44% increase with respect to a 2-D mesh of the same size. That is, in a system with 2 mm $\times$ 2 mm processing elements, the increase in the chip area is less than 4.5%. The area of the network part of a 16×16 2-D BM network shows a 49% increase compared to the 2-D mesh, which is an increase of less than 5% in the entire chip area.

It is noteworthy that the router area is a function of its buffers depth. The depth of the input and output buffers of the routers, in this paper, were set to 2 flits (or 4-flit buffer at each port). A performance analysis in [34], however, showed that a typical router in next generation high-performance CMPs (using 22 nm technology) will need 64-flit buffers at each port, which result in a higher router/switch area. Figure 18 plots the area overhead of the digraph-based NoCs (over a mesh with the same size and buffering capacity) for two network sizes $N = 8 \times 8$ and 16×16 and different buffer depths in a system with 32-bit channels. Based on Fig. 18, a system using 64-flit buffers reduces the area overhead of the 2-D SEM networks to 5–14% and the area overhead of the 2-D BM networks to 9–16%.

6 Summary of results

According to the simulation results reported in Sect. 5, we can conclude that the 2-D digraph-based NoCs have superior performance over the 2-D mesh NoC, since the average hop counts traversed by messages in a 2-D digraph-based network is smaller than that in the equivalent 2-D mesh.

The node degrees of the 2-D digraph-based networks and 2-D mesh networks are the same. However, unlike the 2-D mesh topology, the 2-D digraph-based networks have some links that connect non-adjacent nodes and therefore, they may be longer than links in their 2-D mesh counterparts. This can lead to an increase in the network area and also cause difficulties in link placement. The latter can be solved by a number of efficient VLSI layouts proposed for digraphs (i.e. the layout proposed in [14]).

Simulation results indicate that the power consumption of a 2-D digraph-based network is less than the power dissipated in the equivalent 2-D mesh NoC, for light to medium traffic loads. However, it begins to behave differently near heavy traffic regions where the 2-D mesh network saturates and cannot handle more traffic.

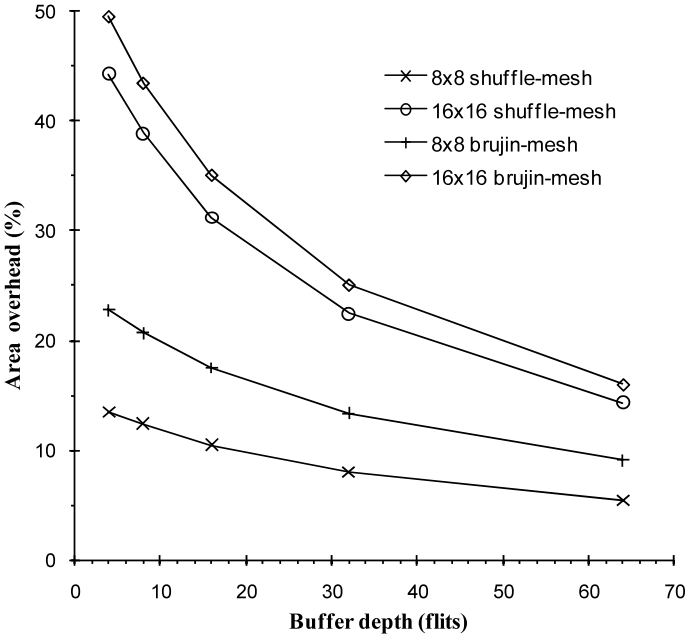


Fig. 18 The area overhead of 8×8 and 16×16 digraph-based mesh NoCs with different buffer depths

According to the simulation results, the 2-D digraph-based tori exhibit even better performance compared to their 2-D digraph-based mesh counterparts. The reason is that the average distance a message travels in the network in a 2-D digraph-based torus is lower than that in the 2-D digraph-based mesh counterpart.

7 Conclusion and future work

The 2-D mesh topology has been widely used in NoC design. However, directed graphs have not been studied yet as the underlying topology for two-dimensional NoCs. This paper introduced two-dimensional digraph-based meshes as attractive alternatives to the mesh topology for NoCs.

Extensive simulation experiments were conducted to assess the network latency and power consumption of the proposed networks. Results showed that the proposed networks improved the performance (in term of average message latency) of the network. The power consumption in the proposed networks was also less than that of the equivalent mesh NoC under light and moderate traffic loads. However, the proposed networks consumed more power in heavy traffic loads as a direct result of the more traffic handled where the equivalent mesh network was saturated and could not take any more traffic load.

Wire lengths and area for the proposed networks were also compared with those in a mesh and it was found that the proposed networks do not add much area especially when compared within a whole system on chip (i.e. IP cores plus NoC).

This study was based on synthetic traffic loads. Future work in this line can consider realistic traffic loads and evaluate the suitability of the 2-D digraph-based networks as the underlying topology for NoCs.

Reconfigurable multi-ring networks were introduced and studied in the past [35–37]. They have desirable properties and are good candidates for on-chip networks as they are inherently low-dimensional and can react dynamically to the network performance requirements. Therefore, proposing a new on-chip network based on the reconfigurable multi-ring architecture and evaluating its performance and power consumption under different working conditions can also be another topic for future research in this line.

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